

Machine_Learning_Proposal

Customer Churn Prediction for ABC Communications Ltd

Machine Learning Internship Programme
AnalystLab Africa

Week 1 Assignment

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1. Project Overview

ABC Communications Ltd is a telecommunications company experiencing customer attrition, commonly referred to as customer churn. Customer churn negatively impacts revenue, increases customer acquisition costs, and reduces long-term customer value.

This proposal outlines the machine learning approach for developing a predictive model capable of identifying customers who are likely to discontinue their services. By predicting churn before it occurs, the company can implement proactive customer retention strategies that improve customer satisfaction and business profitability.

The proposal summarizes the selected machine learning approach, preprocessing strategy, recommended algorithms, evaluation metrics, and the expected business impact of the project.

2. Problem Definition

The primary business problem is the inability of ABC Communications Ltd to identify customers who are at risk of leaving the company.

Without predictive insights, the organization reacts only after customers have already churned, leading to revenue loss and increased operational costs.

The proposed machine learning solution will predict whether a customer is likely to churn using historical customer information.

3. Project Objectives

The objectives of this project are:

- Predict customer churn before customers leave.
- Improve customer retention through proactive intervention.
- Support data-driven business decision-making.
- Reduce customer acquisition costs.
- Increase customer lifetime value.
- Improve customer satisfaction and loyalty.

4. Machine Learning Approach

The project follows the supervised machine learning approach because historical data contains known churn outcomes.

Each customer record includes a label indicating whether the customer stayed or left the company.

The machine learning model will learn relationships between customer characteristics and churn status and use these learned patterns to predict future customer behaviour.

The problem is formulated as a **binary classification** task because the target variable contains two possible outcomes:

- Yes (Customer Churned)
- No (Customer Retained)

5. Dataset Description

The project uses the **Telco Customer Churn Dataset** obtained from Kaggle.

Dataset characteristics include:

Item	Description
Dataset	Telco Customer Churn
Source	Kaggle
Records	7,043
Features	21
Target Variable	Churn
Problem Type	Binary Classification

The dataset contains customer demographic information, subscribed services, contract details, billing information, and customer churn status.

6. Proposed Data Preprocessing Strategy

Before training machine learning models, the dataset will undergo preprocessing to improve data quality and model performance.

The preprocessing strategy includes:

- Removing the customer identifier column (customerID).
- Converting the TotalCharges column to a numerical data type.
- Handling missing values using median imputation.
- Encoding categorical variables using One-Hot Encoding.
- Encoding the target variable into binary values.
- Splitting the dataset into training and testing sets using an 80:20 ratio.

These steps ensure that the dataset is suitable for machine learning algorithms.

7. Recommended Machine Learning Algorithms

Several supervised learning algorithms are suitable for customer churn prediction.

Logistic Regression

A simple and interpretable baseline classification model.

Decision Tree

Creates decision rules that are easy to understand and visualize.

Random Forest

Combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Support Vector Machine (SVM)

Effective for binary classification problems with complex decision boundaries.

XGBoost

A powerful gradient boosting algorithm known for its high predictive performance.

LightGBM

An efficient gradient boosting algorithm optimized for speed and scalability.

These algorithms will be compared to identify the best-performing model.

8. Model Evaluation

To assess model performance, the following evaluation metrics will be used:

Metric	Purpose
Accuracy	Measures overall prediction correctness
Precision	Evaluates the proportion of correctly predicted churn cases
Recall	Measures the model's ability to identify actual churners
F1-Score	Balances precision and recall
ROC-AUC Score	Measures the model's ability to distinguish between classes
Confusion Matrix	Summarizes correct and incorrect predictions

Using multiple evaluation metrics provides a comprehensive assessment of model performance.

9. Expected Business Benefits

Implementing the proposed machine learning solution is expected to provide several benefits to ABC Communications Ltd:

- Early identification of customers at risk of churn.
- Improved customer retention.
- Reduced revenue loss.
- Lower marketing and customer acquisition costs.

- Better customer relationship management.
- Enhanced decision-making through predictive analytics.
- Increased profitability and long-term business sustainability.

10. Potential Challenges

Several challenges may affect model performance:

- Class imbalance between churned and retained customers.
- Missing or inconsistent data.
- Changes in customer behaviour over time.
- Overfitting of complex machine learning models.

These challenges can be mitigated through appropriate preprocessing, feature engineering, cross-validation, and model tuning.

10. Future Work

Future phases of the project will include:

- Training multiple machine learning models.
- Hyperparameter tuning.
- Model comparison.
- Feature importance analysis.
- Model deployment using a web application or API.
- Continuous monitoring and periodic model retraining.

10. Conclusion

This proposal outlines a structured machine learning approach for predicting customer churn at ABC Communications Ltd.

The project has been framed as a supervised binary classification problem using the Telco Customer Churn dataset. Appropriate preprocessing techniques, machine learning algorithms, and evaluation metrics have been identified to support successful model development.

By implementing this solution, the organization can proactively identify customers who are likely to leave, enabling timely retention strategies that improve customer satisfaction, reduce churn, and increase business profitability.